NMM 1411A – Practice Test 2

Note. This practice test contains about twice as many questions as the actual Test 2. The numerical answers are not necessarily "nice", so calculators may be used **for this practice test**. **No solutions will be posted.**

1. (a) Solve the following system of linear equations by using the inverse of the coefficient matrix. Show the inverse matrix you use.

$$\begin{cases} 1.8x - 0.7y = 3.1, \\ 0.9x + 1.4y = 2.6. \end{cases}$$

$AX = B$, $A = \begin{bmatrix} 1.8 & -0.7 \\ 0.9 & 1.4 \end{bmatrix}$, $B = \begin{bmatrix} 3.1 \\ 2.6 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$A^{-1} \cdot A \cdot X = A^{-1} B$, $I = 1$, $X = A^{-1} \cdot B$

$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ c & a \end{bmatrix}$ 0.31746

$= \frac{1}{3.15} \begin{bmatrix} 1.4 & 0.7 \\ 0.9 & 1.8 \end{bmatrix}$

$= \begin{bmatrix} 0.44 & 0.22 \\ 0.29 & 0.57 \end{bmatrix}$

(b) Consider the system

$$\begin{cases} x_1 + 2x_2 - x_3 = b_1, \\ 2x_1 - x_2 + 3x_3 = b_2, \\ 0.5x_1 + 3.5x_2 - 3x_3 = b_3. \end{cases}$$

Find the condition on b_1, b_2, b_3 that guarantees that the system is consistent. Briefly justify your answer using row reduction.

(c) Let

$$D = \operatorname{diag}(1.6, -0.75, 2.5).$$

Compute D^{-2}.

2. (a) Calculate the determinant of

$$A = \begin{bmatrix} 1.3 & 0 & -0.8 & 0 \\ 0 & 2.1 & 0 & 0 \\ 0.6 & 0 & 1.7 & 0 \\ 2.2 & -0.5 & 1.1 & -1.4 \end{bmatrix}.$$

Choose a cofactor expansion that makes the calculation efficient.

(b) Let A be a 3×3 matrix with $\det(A) = -2.4$. Compute

$$\det(1.7A^T).$$

State the determinant properties you use.

(c) Use Cramer's rule to solve the system

$$\begin{cases} x + 1.3y = 2.7, \\ y + 0.8z = 3.2, \\ 1.1x + z = 4.1. \end{cases}$$

3. (a) Solve the following system of linear equations by using the inverse of the coefficient matrix. Show the inverse matrix you use.

$$\begin{cases} 1.4x + 2.1y = 5.3, \\ -0.6x + 1.7y = 1.8. \end{cases}$$

(b) Consider the system

$$\begin{cases} 2x_1 - x_2 + x_3 = b_1, \\ x_1 + 2x_2 + 3x_3 = b_2, \\ -0.25x_1 + 3.25x_2 + 3x_3 = b_3. \end{cases}$$

Find the condition on b_1, b_2, b_3 that guarantees that the system is consistent. Briefly justify your answer using row reduction.

(c) Let

$$D = \mathrm{diag}(-1.2, 0.8, 2.2).$$

Compute D^{-2}.

4. (a) Calculate the determinant of

$$A = \begin{bmatrix} 2.2 & 0 & 0.7 & 0 \\ 0 & -1.6 & 0 & 0 \\ 1.1 & 0 & 2.4 & 0 \\ -0.5 & 1.3 & 0.9 & 1.8 \end{bmatrix}.$$

Choose a cofactor expansion that makes the calculation efficient.

(b) Let A be a 3×3 matrix with $\det(A) = 1.6$. Compute

$$\det(1.3A^T).$$

State the determinant properties you use.

(c) Use Cramer's rule to solve the system

$$\begin{cases} 0.9x + y = 2.4, \\ 1.2y + z = 3.7, \\ x + 0.7z = 3.1. \end{cases}$$